

Lab 04: Chihuahua or Muffin

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Reflection Journal

Lab Objective:

This lab focused on building an image classifier using machine learning techniques to determine whether an image is a chihuahua or a muffin. It utilizes PyTorch to load image data, train a network, and check accuracy on the validation set. The challenge was to push validation accuracy above 80%. The starting model was a fully connected network that achieved 54% accuracy before any changes were made.

Key Concepts Learned

Convolutional Neural Networks (CNNs) detect spatial patterns with filters instead of flattening an image into a flat vector. This process makes CNNs better than neural networks at handling image data. Transfer learning reuses a model that has already been trained on a large dataset. ResNet18 is a CNN pretrained on image data and was used to achieve higher accuracy. Data augmentation can help cut overfitting for small datasets.

Challenges Encountered

The original network ran without errors, but its accuracy was lacking. A Jupyter/Colab working directory issue caused image loading to break; it was fixed by reloading the notebook

and cloning the repo. Overfitting was avoided by freezing the pretrained backbone, adding dropout, and using augmentation.

Insights Gained About Machine Learning & Image Classification

Architecture is important and matters most when training a model. When selecting a base architecture, it is important to clearly define the model's objective to select the appropriate feature extractors and training frameworks. Transfer learning can be a high-leverage tool when working with small datasets. 100% validation accuracy is a good signal, but not proof that the model can generalize. A model could look perfect but be sensitive to new data.

Potential Real-World Applications

CNN-based transfer learning can be used in medical imaging to classify pneumonia on chest X-rays. Gu & Lee (2024) documented a similar approach in which pretrained models converged faster and achieved higher accuracy, recall, precision, and F1 scores than models trained from scratch.

Personal Reflection

This changed my view of CNNs, that more layers or training time would improve accuracy. Proper architecture mattered more when fitting a model. It also helped reinforce a debugging habit. If a model performs poorly, check whether the architecture fits the problem before tuning anything else.

Works Cited

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